CS4782 Final Project Proposal

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Paper Title

DoRA: Weight-Decomposed Low-Rank Adaptation (https://arxiv.org/pdf/2402.09353)

- Authors: Shih-Yang Liu, Chien-Yi Wang, Hongxu Yin, Pavlo Molchanov, Yu-Chiang, Frank Wang, Kwang-Ting Cheng, Min-Hung Chen
- Publication venue: 41st ICML (2024)

Brief summary of the chosen paper

Fine-tuning is a method to further improve machine learning models that have been pre-trained on a general task, by feeding the model a more specific, smaller dataset to make it excel at that particular task. However, traditional full fine-tuning (FT) is very computationally expensive for modern large language models, as it can update many billions of parameters. This gives rise to parameter-efficient fine-tuning (PEFT) techniques, which can achieve similar performance to FT while only updating a fraction of the parameters. A previously proposed PEFT method, Low-Rank Adaptation (LoRA), gained popularity due to relatively good performance and avoiding additional inference costs, but still has a significant accuracy gap with FT. The authors propose Weight-Decomposed Low-Rank Adaptation (DoRA), which builds upon LoRA but decomposes the learned weight matrix into a magnitude vector and a direction matrix, applying LoRA updates to only the direction matrix. This led to much higher accuracy comparable to FT while attaining even greater efficiency than LoRA.

Brief justification of why you chose this paper

We chose this paper because we thought it showed a lot of promise in providing an efficient way to tune LLMs to particular tasks. We thought it would be interesting to have an LLM expert in competitive math, as we all have done competitive math in the past.

Prove that the data exists/is accessible

We plan to use several publicly available math competition datasets, primarily sourced from competitions such as AIME, USAMO, and others hosted on the Art of Problem Solving (AoPS) website. Conveniently, multiple open-source datasets have already assembled and organized these problems, and more competition problems:

- **HARP (Human-Annotated Reasoning Problems)**: https://arxiv.org/abs/2412.08819
    - Various math competition problems with human-annotated reasoning traces.

- **Omni-Math:** https://huggingface.co/datasets/KbsdJames/Omni-MATH
    - A dataset of ~4K Olympiad-level math problems for evaluating advanced mathematical reasoning.
- **MATH**: https://huggingface.co/datasets/DigitalLearningGmbH/MATH-lighteval
    - A dataset of ~12.5K competition math problems with step-by-step solutions for training and benchmarking math reasoning models.
- **Math Datasets 100k**: https://huggingface.co/datasets/weijiezz/math-datasets-100k
    - A collection of 100k math problems aggregated from multiple competition datasets. It also includes a clean test split containing recent exams such as AIME25.

We will cross-check these datasets for overlapping problems and manually filter them to ensure the final dataset remains clean.

- If we require any larger-scale training data, we can additionally use **OpenMathInstruct-2**: https://huggingface.co/datasets/nvidia/OpenMathInstruct-2
    - A much larger instruction-style math dataset suitable for more training.

Result Selection

(Tell us which results you want to replicate)
We want to replicate the main result from the paper, showing that by using certain decomposition techniques, DoRA consistently outperforms LoRA with the same parameter budget. We aim to apply this to several new math based datasets, as shown above. In addition, we hope to replicate the result in the paper showing that DoRA even with half rank performs similarly to LoRA, showing significantly better parameter efficiency.g

Re-implementation Plan

(Describe architecture, method, and metrics)

We are building this project in Python and will load a base model (particularly Llama) using Hugging Face's transformers library. We will then write DoRA configuration objects that will handle the weight decomposition, and use the *peft* library to inject our custom DoRA into the base model. This requires us to redefine forward pass layers. After training normally, we can then utilize the datasets library in order to ingest the math datasets we have selected in order to obtain our training data. In order to assess our results, our primary metric will be accuracy on a clean test set which can be recent math competition problems. We will compare our DoRA implementation against the baseline DoRA logic defined in *peft*.

Details about how much compute and time are required to replicate results (make sure that the amount of compute needed is feasible)

We will be utilizing Google Colab's NVIDIA T4 GPU (16GM VRAM) to eliminate the need for local hardware. In order to ensure that we are able to run a 7B or 8B parameter model (like Llama) we can utilize the *bitsandbytes* library which leaves us enough memory to train our custom magnitude vectors and LoRA adapters.